

Experiment No.12
FLIP-FLOPS (D and T)

Course Code: CPE 013

Program:

Course Title: Logic Circuits and Design

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Section: 22S2

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1. Objective(s):

This activity aims to apply the functions of D and T flip-flop

2. Intended Learning Outcomes (ILOs):

The students shall be able to:

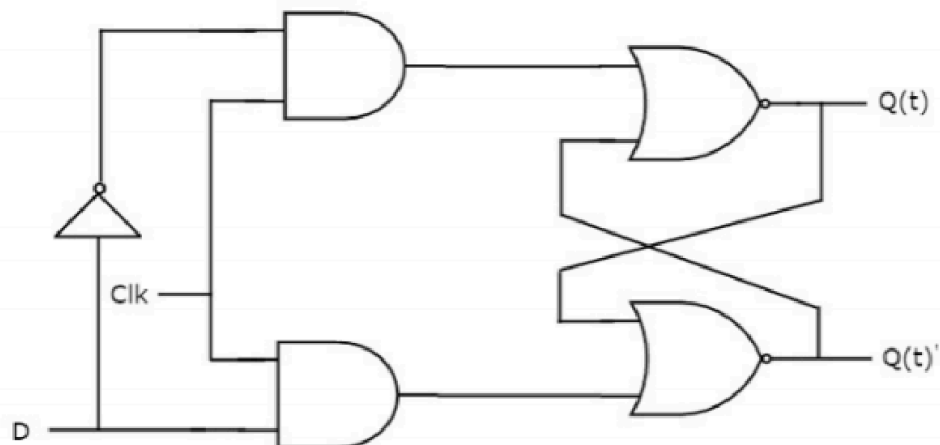
2.1 Apply the D and T flip-flop circuits to the Logically and TinkerCAD application in generating circuit.

2.2 Show the output of the simulated circuit using TinkerCAD simulation application.

3. Discussion:

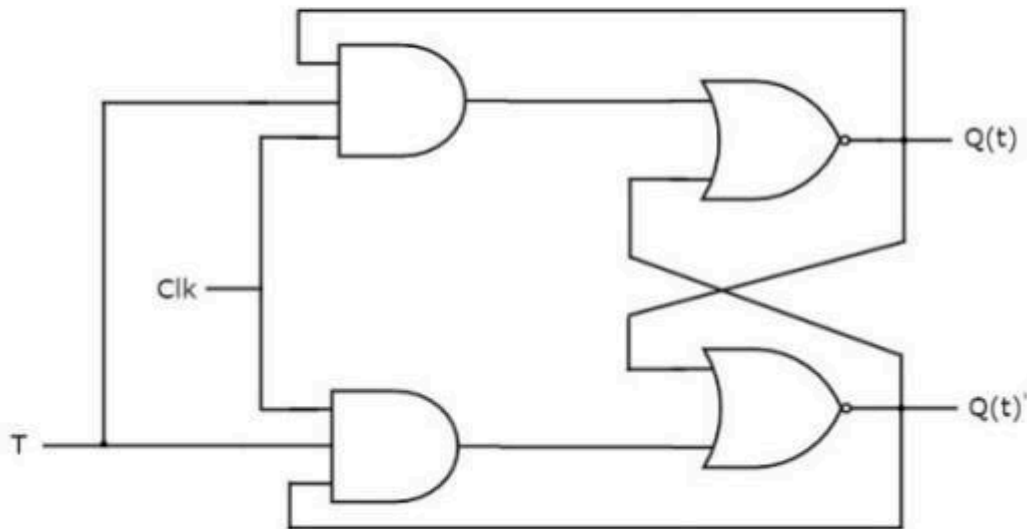
D Flip-Flop

D flip-flop operates with only positive clock transitions or negative clock transitions. Whereas, D latch operates with enable signal. That means, the output of D flip-flop is insensitive to the changes in the input, D except for active transition of the clock signal. The **circuit diagram** of D flip-flop is shown in the following figure.



T Flip-Flop

T flip-flop is the simplified version of JK flip-flop. It is obtained by connecting the same input 'T' to both inputs of JK flip-flop. It operates with only positive clock transitions or negative clock transitions. The **circuit diagram** of T flip-flop is shown in the following figure.



Source: https://www.tutorialspoint.com/digital_circuits/digital_circuits_flip_flops.htm

4. Materials:

- Personal Computer
- Internet connection

Simulation Tools:

- o TinkerCAD
- o Logicly

5. Procedure:

1. Construct and simulate the operation of each flip flop in Logic.ly using basic logic gates and plot the output in a truth table.
2. Research on the internal circuitry and Pin Configurations of all Flip flops ICs.
3. Create a truth table, show solution using k-map, draw the circuit diagram of the following problems:

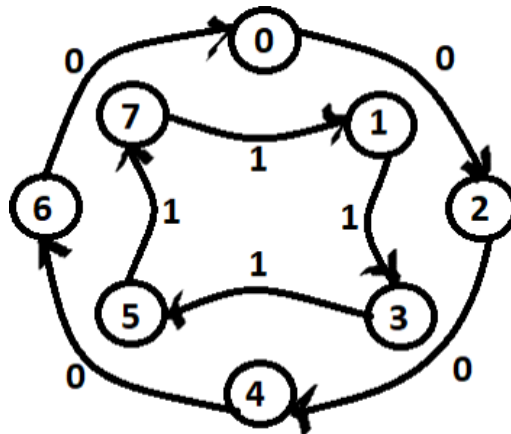
TRUTH TABLE

Input (T)	Present State Q(t)	Next State Q(t+1)	Q(t+1)\$Operation
0	0	0	Hold(No change)
0	1	1	Hold(No change)
1	0	1	Toggle(Invert)
1	1	0	Toggle (Invert)

KMAP

T\Q(t)	0	1
0	0	1
1	1	0

- Design a 2-bit counter using D flip-flop displayed in LEDs
- Design a 2- bit counter using T flip-flop displayed in LEDs

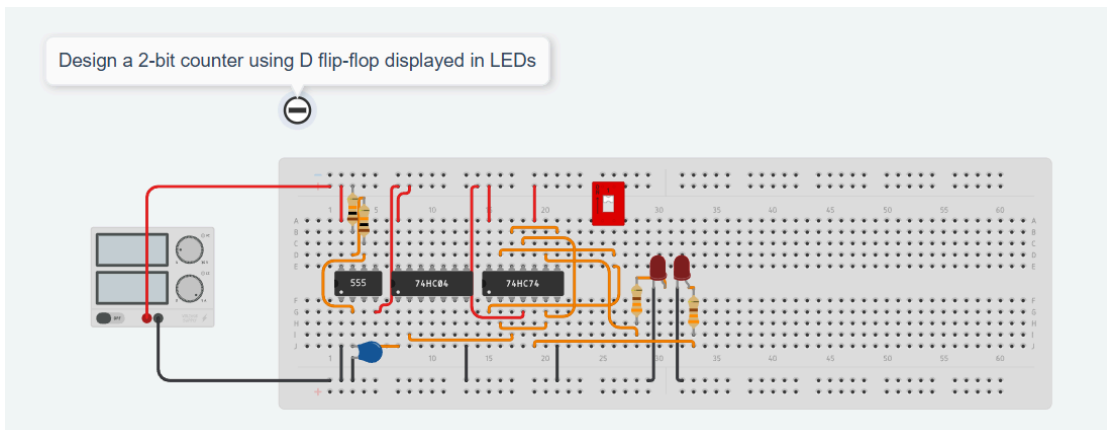
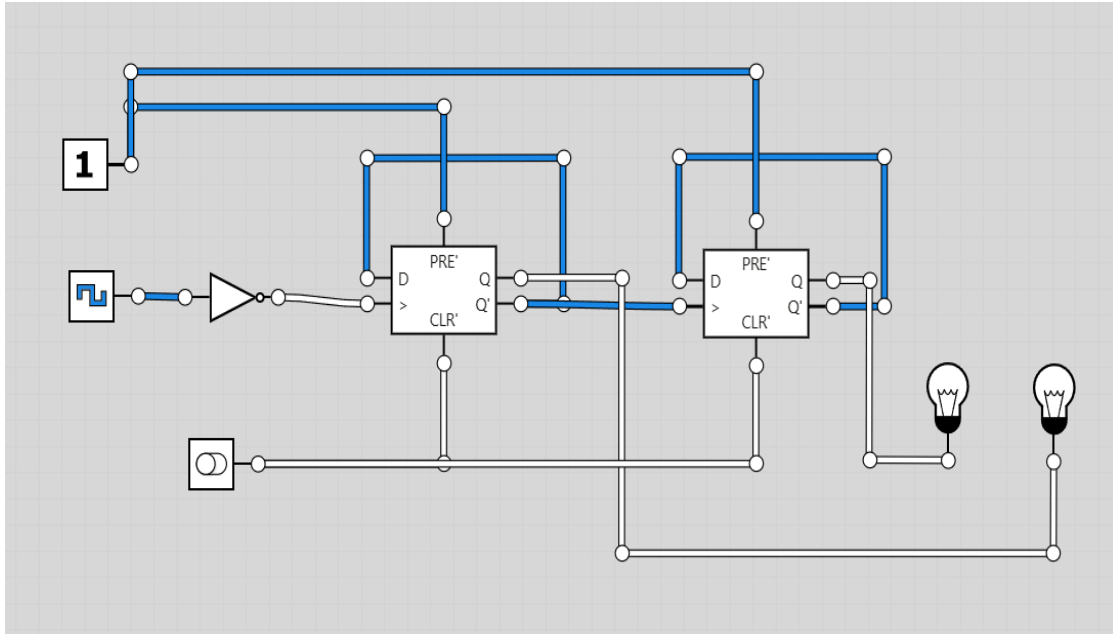


- Construct and simulate the in TinkerCAD by connecting the inputs, wires, electronic components, and output correctly following the pin configuration of the IC.
- After the pasting an output screenshot and writing down the link for the uploaded video recording below, write the conclusion, and answer the questions on section 8.

6. Data and Results:

Put here the State Diagram and your solution for the given problem in D and T Flip-flop.

a) Design a 2-bit counter using D flip-flop displayed in LEDs



TRUTH TABLE

CLK	D	Q(Present state)	Q'(Present state)	Q(Next state)	Q'(Next state)
↑	0	0	1	0	1
↑	0	1	0	0	1
↑	1	0	1	1	0
↑	1	1	0	1	0

K-MAP

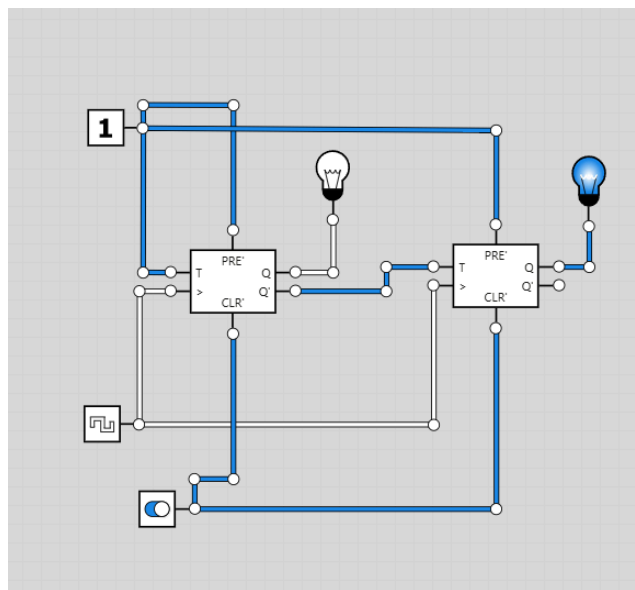
D0

Q1/Q0	0	1
0	1	0
1	1	0

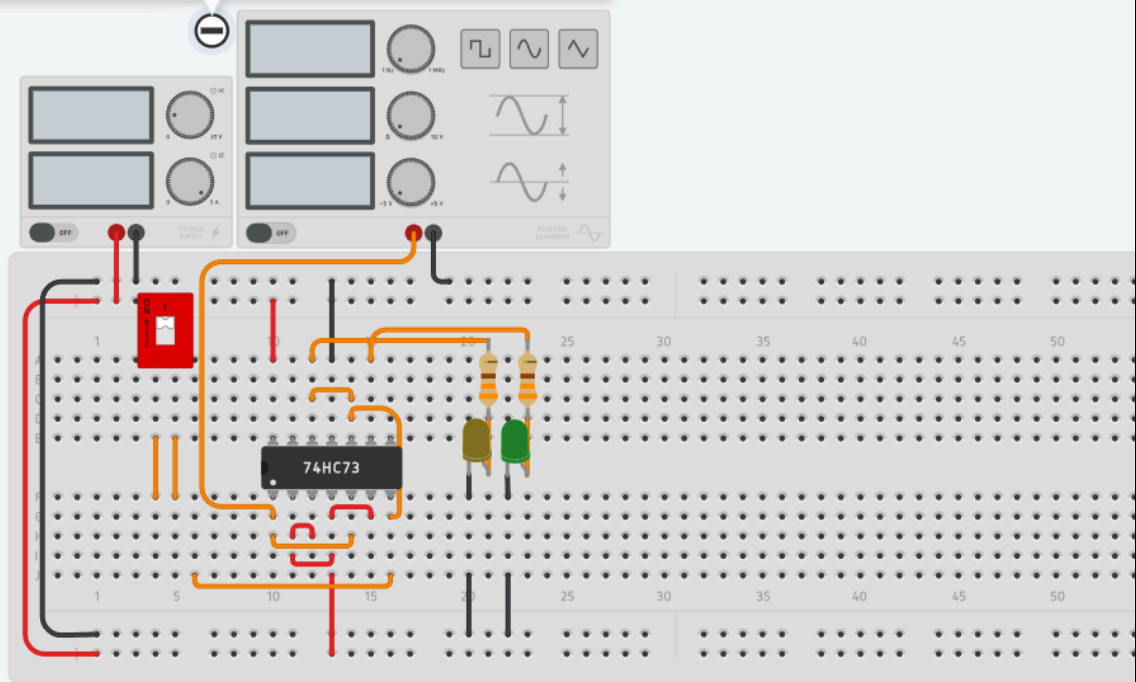
D1

Q1/Q0	0	1
0	0	1
1	1	0

b) Design a 2-bit counter using T flip-flop displayed in LEDs



Design a 2-bit counter using T flip-flop displayed in LEDs



TRUTH TABLE

CLK	T	Q(Present state)	Q'(Present state)	Q(Next state)	Q'(Next state)
↑	1	0	1	0	1
↑	1	1	0	1	0
↑	1	0	1	1	0
↑	1	1	0	0	1

7. Final Output:

Screenshot or print screen of TinkerCAD application different output simulations (5 points deductions for only one output in every required circuit). Also, upload a video of the simulated circuit naming or talking about the shown input combinations. Failure to comply with each instruction will result in 5 points deductions. Please include the recorded simulation video in this section and write at least one

sentence definition or label for each figure or image.

TINKERCAD LINK:

<https://www.tinkercad.com/things/6uVaNHjePN6-2bit-counter-using-t-and-d-flip-flops>

VIDEO DEMO: 2BITCOUNTERSFLIPFLOP

8. Conclusion:

Write a one paragraph (at least 10 sentences) Conclusion for this activity.

The experiment successfully demonstrated and compared 2-bit synchronous down-counters using both D-type and T-type flip-flops. While the counter's core function was achieved, the T flip-flop implementation faced a key challenge: confusion with the pin configuration during the necessary JK to T-type conversion. Despite this initial technical hurdle and delay, the team overcame the wiring difficulty. The final circuits performed as predicted, allowing for a clear comparison between the D and T flip-flop designs.

9. Questions:

1. What have you observed in the block diagram of D Flip-flop? What is the application of D flip-flop?

In the block diagram, we observe a single data input (D) and a clock input, where the output Q simply adopts the state of D at the triggering clock edge. Because of this "data transparent" behavior, its primary applications include data storage (registers), memory units, and delay elements in digital systems.

What have you observed in the block diagram of T Flip-flop? What are the application of T flip-flop?

The T flip-flop toggles (complements) its output (Q) when the toggle input (T) is high during a clock pulse, otherwise it holds its state. This characteristic is ideal for frequency division and binary counters, as it simplifies state transition logic.

10. Assessment (Rubric for Laboratory Performance):

Laboratory Performance Rubric (2)					
Criteria	Ratings			Pts	
Manipulative Skills	3 pts Proficient Student always demonstrate needed skills.	2 pts Acceptable Student occasionally demonstrate needed skills.	1 pts Beginner Student do not demonstrate needed skills.	3 pts	
Experimental Set - up	3 pts Proficient Student are able to set-up the material with minimum supervision.	2 pts Acceptable Student are able to set-up the materials with supervision.	1 pts Beginner Student are unable to set-up the materials.	3 pts	
Process Skills	3 pts Proficient Student always demonstrate targeted process skills.	2 pts Acceptable Student occasionally demonstrate targeted process skills.	1 pts Beginner Student do not demonstrate targeted process skills.	3 pts	
Application of Technical Writing Skills	3 pts Proficient Correct spelling, grammar, and sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	2 pts Acceptable Some minor incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	1 pts Beginner Many incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	3 pts	
Time Management / Conduct of Experiment	3 pts Proficient Student finish ahead of time with complete data and time to revise data.	2 pts Acceptable Student finish on time with incomplete data.	1 pts Beginner Student do not finish on time with incomplete data.	3 pts	
Complete Requirements of Laboratory Report	3 pts Proficient Student submitted a complete requirements set by the teacher.	2 pts Acceptable Student submitted an almost complete Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student submitted a lot of missing requirements in the Laboratory Report according to requirements set by the teacher.	3 pts	
Follow the Correct Format of Laboratory Report	3 pts Proficient Student submitted a correct format of Laboratory Report according to requirements set by the teacher.	2 pts Acceptable Student submitted an almost correct format of Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student did not follow at all the format of Laboratory Report according to requirements set by the teacher.	3 pts	
Ability to do independent work	3 pts Proficient Student do not need to be supervised by the teacher.	2 pts Acceptable Student require occasional supervision by the teacher.	1 pts Beginner Student require supervision by the teacher.	3 pts	
Total Points: 24					